

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

Technology Executives & Network Infrastructure Leaders

17%

Global Traffic Growth

3.3x

Starlink Surge

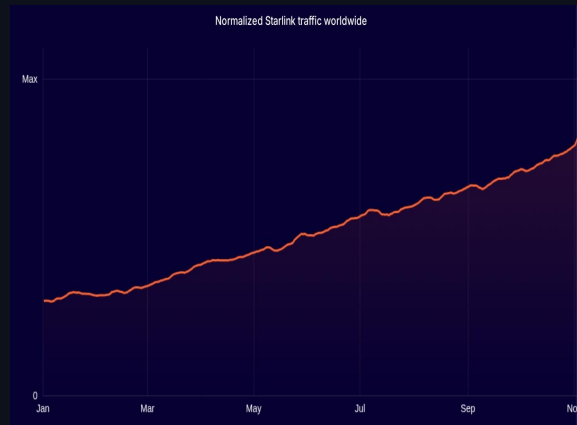
17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Year-over-Year Global
Internet Traffic Growth

3.3x

SpaceX Starlink Traffic Surge
in 2024



- Global Internet traffic grew by 17% in 2024, continuing the upward trend driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation.
- SpaceX Starlink traffic saw dramatic 3.3x growth during 2024, continuing to bring satellite Internet connectivity to previously underserved locations.

Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background. Rankings based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

89%

Google Search Engine
Referral Share

33%

iOS Mobile Traffic Share

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



2% →

13%

Post-Quantum TLS 1.3

21%

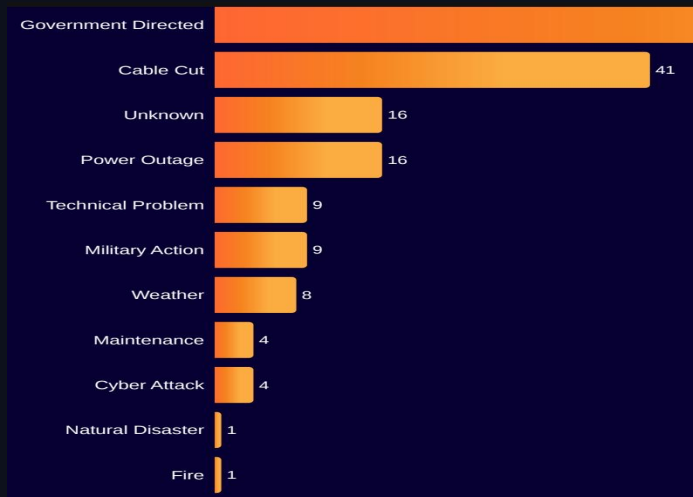
HTTP/3 Adoption

Top API Client Languages

Language	Share
Go	~12%
Python	9.6%
Java	7.4%
NET	3.6%

- Post-quantum encryption surged after Google Chrome 124 enabled it by default on April 17, 2024. Mozilla Firefox rolled it out, and Apple Safari began initial testing.
- HTTP/3 runs on QUIC, providing faster connections, packet-loss mitigation, and encryption by default.

225 Major Disruptions: Government Shutdowns Lead Outage Causes



Government Directed

Cable Cut

Unknown

Power Outage

Technical Problem

Military Action

Weather

Maintenance

Cyber Attack

Natural Disaster



41

16

16

9

9

8

4

4

1

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed globally in 2024, accounting for more than half of all incidents. Cable cuts (41) and power outages (16) followed as significant contributors.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated

68%

Bot Traffic from Top
10 Countries

4.3%

Malicious Emails

43%

Malicious Emails with
Deceptive Links

Key Threat Findings

- Gambling and Games sector was the most attacked industry category.
- Log4j (CVE-2021-44228) continues to be actively targeted more than two years after disclosure.
- RPKI valid routes increased by 6.4%, improving Internet routing security.

Email Security Threats

- 4.3% of emails processed were malicious.
- Among malicious emails, 43% contained a deceptive link.
- The .bar TLD originated the largest share of malicious and spam email observed.

Five Takeaways Shaping the 2025 Internet Outlook

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1 Traffic Growth Trajectory

Global Internet traffic grew 17% year-over-year, with Starlink traffic surging 3.3x as satellite connectivity expanded to underserved regions.

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2 Post-Quantum Momentum

Post-quantum encryption adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024.

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3 Connectivity Fragility

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed, followed by cable cuts (41) and power outages (16).

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4 Persistent Security Threats

6.5% of all traffic was mitigated, Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure.

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5 Evolving Developer Ecosystem

Go surpassed other languages to become the most popular API client language, capturing roughly 12% of automated API traffic.